Project: Worked examples
Title: Lateral-torsional buckling and N+M interaction (EN 1993-1-1)
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1. Scope and method

Lateral-torsional buckling (6.3.2), flexural buckling (6.3.1) and the interaction of axial compression with bending (6.3.3) for a uniform doubly-symmetric I-section member on fork supports, class 1 to 3: M_cr, λ̄_LT, χ_LT and M_b,Rd; χ_y, χ_z and N_b,Rd; the Annex B (or Annex A) interaction factors and both interaction ratios, with the governing check.

WHAT IT CALCULATES	RELATION
Critical moment (NCCI SN003 form)	$M_{cr} = C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 \cdot I_w/I_z + (kL)^2 \cdot G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2)} - C_2 z_g]$; $C_1 = 1.88 - 1.40\psi + 0.52\psi^2 \le 2.70$ (end moments, k = 1) unless entered
LTB slenderness	$\bar{\lambda}_{LT} = \sqrt{(W_y \cdot f_y/M_{cr})}$, $W_y = W_{pl,y}$ (class 1, 2) or $W_{el,y}$ (class 3)
Lateral-torsional buckling, general case (6.3.2.2)	$\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - 0.2) + \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \bar{\lambda}_{LT}^2)}) \le 1$
Lateral-torsional buckling, rolled sections (6.3.2.3)	$\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - \bar{\lambda}_{LT,0}) + \beta \cdot \bar{\lambda}_{LT}^2]$; $\chi_{LT} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \beta \cdot \bar{\lambda}_{LT}^2)}) \le 1, \le 1/\bar{\lambda}_{LT}^2$; optional $\chi_{LT,mod} = \chi_{LT}/f \le 1, \le 1/\bar{\lambda}_{LT}^2$ (eq. 6.58), $f = 1 - 0.5(1 - k_c)[1 - 2(\bar{\lambda}_{LT} - 0.8)^2] \le 1$
Buckling resistance in bending	$M_{b,Rd} = \chi_{LT} \cdot W_y \cdot f_y/\gamma_{M1}$
Flexural buckling	$N_{cr} = \pi^2 EI/L_{cr}^2$, $\bar{\lambda} = \sqrt{(A \cdot f_y/N_{cr})}$, $\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$, $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$, $N_{b,Rd} = \chi \cdot A \cdot f_y/\gamma_{M1}$; the curve of each axis from Table 6.2 (section type, h/b, t_f; $f_y \le 420$ MPa) or chosen
Interaction (6.3.3(4))	$N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1}) + k_{yy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{yz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$; $N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1}) + k_{zy} \cdot M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1}) + k_{zz} \cdot M_{z,Ed}/(M_{z,Rk}/\gamma_{M1}) \le 1$
Factors k_{ij}	Annex B from the slenderness, $N_{Ed}/(\chi \cdot N_{Rk}/\gamma_{M1})$ and C_m ($C_m = 0.6 + 0.4\psi \ge 0.4$ for a linear diagram, or entered); Annex A from the μ, w, n_{pl}, a_{LT} and C_{ij} expressions (class 1, 2: plastic column of Table A.1; class 3: elastic column, without the $0.6\sqrt{(w_i/w_j)}$ factors of k_{yz}, k_{zy})

Basis: Buckling curves; slenderness for flexural buckling[1](6.3.1.2, 6.3.1.3); Lateral-torsional buckling curves: general case[1](6.3.2.2); Lateral-torsional buckling curves for rolled sections and equivalent welded sections[1](6.3.2.3); Uniform members in bending and axial compression[1](6.3.3); Interaction factors k_ij: methods 1 and 2[1](Annex A, Annex B); Elastic critical moment of a doubly-symmetric member[2](Mcr expression); Stability of members (label only, see the edition note)[1](8.3).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
edition	EN 1993-1-1 edition (2005+A1:2014 until the 2022 edition is adopted; the 2022 choice changes only the label, see the sheet note)	2005+A1:2014		spec: BS EN 1993-1-1:2005+A1:2014 with the UK National Annex	any
ltb_{method}	χ_{LT} method	6.3.2.3 (rolled and welded sections)		assumption: 6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower M_b,Rd.	any
annex	Interaction factors	Annex B (method 2)		assumption: Annex B (method 2) chosen; Annex A can be selected for comparison.	any
torsion	Member behaviour for Annex B	susceptible to torsional deformation (Table B.2)		assumption: An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2).	any
f_y	Yield strength (for the thickness)	345	MPa	spec: BS EN 10025-2 S355, 16 mm < t ≤ 40 mm (flange 19.6 mm)	235 … 460 MPa
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 cl. 3.2.6	190 … 220 GPa
G	Shear modulus	81	GPa	spec: BS EN 1993-1-1 cl. 3.2.6: G = E/(2(1 + ν)) ≈ 81 GPa	70 … 90 GPa
γ_{M1}	Partial factor γ_{M1} (member buckling)	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.25
$section_{class}$	Cross-section class (class 4 needs effective properties and is outside this sheet)	1		assumption: Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2).	any
$section_{type}$	Section type (selects the buckling curve of Table 6.4 / 6.5)	rolled I		spec: BS EN 10365 UB 457×191×98, hot-rolled	any
h	Overall depth h	467.2	mm	spec: BS EN 10365 UB 457×191×98	50 … 2500 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
b	Flange width b	192.8	mm	spec: BS EN 10365 UB 457×191×98	30 … 1000 mm
t_f	Flange thickness t_f (S1: t_f; selects the flexural buckling curves of Table 6.2)	19.6	mm	spec: BS EN 10365 UB 457×191×98	2 … 150 mm
A	Area A	125.265	cm²	reference: S1 section sheet SEC-01, UB 457×191×98 (A = 125.265 cm²)	5 … 3000 cm²
I_y	Second moment of area, major axis I_y (S1: I_{xx})	45728	cm⁴	reference: S1 section sheet SEC-01 (I_xx = 45727.99 cm⁴)	50 … 5000000 cm⁴
I_z	Second moment of area, minor axis I_z (S1: I_{yy})	2347.02	cm⁴	reference: S1 section sheet SEC-01 (I_yy = 2347.02 cm⁴)	5 … 500000 cm⁴
I_t	Torsion constant I_t (S1: J)	121.287	cm⁴	reference: S1 section sheet SEC-01 (J = 121.287 cm⁴)	0.1 … 100000 cm⁴
I_w	Warping constant I_w	1.17554	dm⁶	reference: S1 section sheet SEC-01 (I_w = 1.17554 dm⁶)	0.00001 … 100000 dm⁶
$W_{pl,y}$	Plastic modulus, major axis (S1: $W_{pl,x}$)	2232.46	cm³	reference: S1 section sheet SEC-01 (W_pl,x = 2232.46 cm³)	1 … 100000 cm³
$W_{el,y}$	Elastic modulus, major axis (S1: $W_{el,x}$)	1957.53	cm³	reference: S1 section sheet SEC-01 (W_el,x = 1957.53 cm³)	1 … 100000 cm³
$W_{pl,z}$	Plastic modulus, minor axis (S1: $W_{pl,y}$)	378.904	cm³	reference: S1 section sheet SEC-01 (W_pl,y = 378.904 cm³)	0.5 … 50000 cm³
$W_{el,z}$	Elastic modulus, minor axis (S1: $W_{el,y}$)	243.467	cm³	reference: S1 section sheet SEC-01 (W_el,y = 243.467 cm³)	0.5 … 50000 cm³
L	Length between lateral restraints (fork supports)	5	m	drawing: FR-BM-12 rev B	0.1 … 100 m
k	Effective length factor k (end rotation about the minor axis)	1		assumption: Fork supports at the lateral restraints (no end restraint of the minor-axis rotation).	0.5 … 1
k_w	Warping factor k_w	1		assumption: Free to warp at the restraints.	0.5 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
z_g	Load height z_g above the shear centre, towards the compression flange (positive destabilises)	0	mm	assumption: End-moment loading: no transverse load, so the load height does not enter.	-1000 … 1000 mm
C_2	C_2 (0 for end moments; enter it for transverse loads)	0		assumption: C_2 is 0 for end-moment loading.	0 … 1.5
ltb_{curve}	Buckling curve for χ_{LT}	auto (Table 6.4 / 6.5)		assumption: Curve from Table 6.5 for a rolled I with h/b > 2.	any
use_f	Apply the f modification to χ_{LT} (6.3.2.3(2))	yes		assumption: The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula.	any
α_{a0}	Imperfection factor α, curve a0 (recommended 0.13)	0.13		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
α_a	Imperfection factor α, curve a (recommended 0.21)	0.21		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
α_b	Imperfection factor α, curve b (recommended 0.34)	0.34		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
α_c	Imperfection factor α, curve c (recommended 0.49)	0.49		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
α_d	Imperfection factor α, curve d (recommended 0.76)	0.76		spec: EN 1993-1-1 Table 6.1, recommended value	0.05 … 1
λ_{LT0}	$\bar{\lambda}_{LT,0}$ for 6.3.2.3 (recommended 0.4)	0.4		spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value	0.2 … 0.5
β_{LT}	β for 6.3.2.3 (recommended 0.75)	0.75		spec: EN 1993-1-1 cl. 6.3.2.3(1), recommended value	0.5 … 1
$moment_{mode}$	Moment diagram given as	end moments		assumption: The moments are the member end moments.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_{Ed}	Design axial force, TENSION positive and compression negative (as S2 N_{e1} and S3; 6.3.3 applies to compression)	-400	kN	reference: S2 frame sheet FR-01, member 2, ULS combination (compression negative)	-100000 … 0 kN
$M_{y,A}$	Moment at end A about y-y, as an ordinate of the bending-moment diagram (end-moment mode): the same sign at both ends is single curvature, opposite signs double curvature. Not the end actions of a stiffness analysis, which flip sign at one end	340	kN·m	reference: S2 frame sheet FR-01, member 2, end A	-100000 … 100000 kN·m
$M_{y,B}$	Moment at end B about y-y, same convention as A: a bending-moment-diagram ordinate, the same sign as A for single curvature (end-moment mode)	170	kN·m	reference: S2 frame sheet FR-01, member 2, end B (same side: single curvature)	-100000 … 100000 kN·m
$M_{z,Ed}$	Greatest design moment about z-z	6	kN·m	assumption: Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate).	0 … 100000 kN·m
ψ_z	ψ of the M_z diagram	1		assumption: Uniform minor-axis moment (conservative).	-1 … 1
$L_{cr,y}$	Buckling length about y-y	5	m	drawing: FR-BM-12 rev B	0.1 … 100 m
$L_{cr,z}$	Buckling length about z-z	5	m	drawing: FR-BM-12 rev B	0.1 … 100 m
$curve_y$	Buckling curve about y-y (auto: Table 6.2 from the section type, h/b and t_f, for f_y up to 420 MPa)	auto (Table 6.2)		spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about y-y gives curve a	any
$curve_z$	Buckling curve about z-z (auto: Table 6.2 from the section type, h/b and t_f, for f_y up to 420 MPa)	auto (Table 6.2)		spec: EN 1993-1-1 Table 6.2: rolled I, h/b > 1.2, t_f ≤ 40 mm, buckling about z-z gives curve b	any

3. Assumptions

A1. Uniform doubly-symmetric I-section member on fork supports (free to warp and to rotate about the minor axis at the ends, restrained against twist); no mono-symmetric or tapered members, no cantilevers,

no continuous-restraint effects. (Scope of S7; k and k_w below 1 model partial end restraint of the minor-axis rotation and of warping.)
A2. Class 1, 2 or 3 as stated by the user; class 4 effective properties are outside this sheet. (6.2.2.5 is not implemented.)
A3. C_1 from the ψ formula applies to end-moment diagrams with k = 1; for transverse loads, k ≠ 1 or other diagrams the user enters C_1, C_2 (and k_c, C_m) from the source in use. With C_1 entered the sheet blocks until k_c (when the f modification is used) and C_{my} (Annex B) are entered too, unless the diagram is given as end moments: ψ is then a real, entered moment shape. (No table of C_1 or C_2 values is shipped (reference records only).)
A4. The imperfection factors, β and $\bar{\lambda}_{LT,0}$ default to the recommended values and are a user-verifiable factor set; the National Annex in use may differ. (Record kind ec3.ltb-factors.)
A5. The 2022 edition choice changes the label printed on the sheet only, by decision: the arithmetic implemented is that of EN 1993-1-1:2005+A1:2014, the differences between the editions are not implemented (they cannot be stated with certainty), and the clause numbers in the 2022 edition differ and must be checked by the user. (The sheet says so (warning `edition`).)
A6. The flexural buckling curves selected automatically are those of Table 6.2 for S 235 to S 420 (rolled I: a, b for h/b > 1.2 and t_f ≤ 40 mm, b, c above 40 mm or h/b ≤ 1.2 up to 100 mm, d, d beyond; welded I: b, c up to 40 mm, c, d above); for f_y above 420 MPa or a combination the table does not cover the curves are chosen by hand. (Table 6.2 is a small selection rule and is not reproduced beyond it.)
A7. The Annex A and Annex B expressions are implemented as published; their coefficients are to be checked against the owner's copy of the standard. (Interaction factors for diagrams other than linear are entered, not derived.)
A8. $\text{ltb}_{\text{method}}$ (χ_{LT} method) is taken as 6.3.2.3 (rolled and welded sections). (6.3.2.3 chosen for a hot-rolled UB; 6.3.2.2 is the alternative and gives a lower $M_{b,Rd}$.)
A9. annex (Interaction factors) is taken as Annex B (method 2). (Annex B (method 2) chosen; Annex A can be selected for comparison.)
A10. torsion (Member behaviour for Annex B) is taken as susceptible to torsional deformation (Table B.2). (An open I-section with no torsional restraint along its length is susceptible to torsional deformation (Table B.2).)
A11. $\text{section}_{\text{class}}$ (Cross-section class (class 4 needs effective properties and is outside this sheet)) is taken as 1. (Class 1 assumed for S355 UB 457×191×98 under N + M; the engineer confirms the classification (EN 1993-1-1 Table 5.2).)
A12. k (Effective length factor k (end rotation about the minor axis)) is taken as 1. (Fork supports at the lateral restraints (no end restraint of the minor-axis rotation).)
A13. k_w (Warping factor k_w) is taken as 1. (Free to warp at the restraints.)
A14. z_g (Load height z_g above the shear centre, towards the compression flange (positive destabilises)) is taken as 0 mm. (End-moment loading: no transverse load, so the load height does not enter.)
A15. C_2 (C_2 (0 for end moments; enter it for transverse loads)) is taken as 0. (C_2 is 0 for end-moment loading.)
A16. $\text{ltb}_{\text{curve}}$ (Buckling curve for χ_{LT}) is taken as auto (Table 6.4 / 6.5). (Curve from Table 6.5 for a rolled I with h/b > 2.)
A17. use_f (Apply the f modification to χ_{LT} (6.3.2.3(2))) is taken as yes. (The f modification of 6.3.2.3(2) is permitted; k_c from the ψ formula.)
A18. $\text{moment}_{\text{mode}}$ (Moment diagram given as) is taken as end moments. (The moments are the member end moments.)
A19. $M_{z,Ed}$ (Greatest design moment about z-z) is taken as 6 kN·m. (Small minor-axis moment from the cladding-rail eccentricity (engineer's estimate).)
A20. ψ_z (ψ of the M_z diagram) is taken as 1. (Uniform minor-axis moment (conservative).)
A21. Flexural buckling curves a (y-y) and b (z-z) from Table 6.2: rolled I-section, h/b = 2.4232, t_f = 19.6 mm. (Automatic selection; choose the curves by hand to override.)

4. Working

[36] $M_{y,Ed} = \max(|M_A|, |M_B|)$ [1](6.3.2)
$= \max(|340|, |170|)\ kN·m$
$= 340\ kN·m$
Greatest design moment about y-y

[37] $\psi_y = \psi = M_{small}/M_{large}$ (same sign: single curvature) [3](B.3)
$= 170 / 340$
$= 0.5$
Ratio of the end moments, taken as ordinates of the bending-moment diagram (same sign at both ends: single curvature)

[38] $C_1 = 1.88 - 1.40\psi + 0.52\psi^2 \le 2.70$ [2](/ ENV 1993-1-1 Annex F)
$= \psi = 0.5$
$= 1.31$
C_1 for end moments (derived)

[39] $Mcr = C_1 \cdot \pi^2 E \cdot I_z/(kL)^2 \cdot [\sqrt{((k/k_w)^2 I_w/I_z + (kL)^2 G \cdot I_t/(\pi^2 E \cdot I_z) + (C_2 z_g)^2)} - C_2 z_g]$ [2]
$= 1.31 \times \pi^2 \times 210000 \times 23470000 / (5000)^2 \times [\sqrt{((1/1)^2 \times 1175500000000/23470000 + (5000)^2 \times 81}$
$1212900 / (\pi^2 \times 210000 \times 23470000) + (0 \times 0)^2) - 0]$
$= 808.38\ kN·m$
Elastic critical moment for lateral-torsional buckling

[40] $W_y = W_{pl,y}$ (class 1, 2) [1](6.3.2.2(1))
$=$ plastic
$= 2232.5\ cm^3$
Section modulus used for bending about y

[41] $\lambda_{LT} = \sqrt{(W_y \cdot f_y/M_{cr})}$ [1](6.3.2.2(1))
$= \sqrt{(2232.5\ cm^3 \times 345\ MPa / 808.38\ kN·m)}$
$= 0.9761$
Non-dimensional slenderness for lateral-torsional buckling

[48] α_{LT} = imperfection factor, curve c [1](6.3.2.3(1), Table 6.5)
$=$ Table 6.5, rolled, h/b = 2.4232
$= 0.49$
Imperfection factor for lateral-torsional buckling

[51] $\Phi_{LT} = 0.5[1 + \alpha_{LT}(\bar{\lambda}_{LT} - \bar{\lambda}_{LT,0}) + \beta \cdot \bar{\lambda}_{LT}^2]$ [1](6.3.2.3(1))
$= 0.5 \times [1 + 0.49 \times (0.9761 - 0.4) + 0.75 \times 0.9761^2]$
$= 0.99843$
Φ_{LT}

[52] $\chi_{LT,0} = 1/(\Phi_{LT} + \sqrt{(\Phi_{LT}^2 - \beta \cdot \bar{\lambda}_{LT}^2)}) \le 1, \le 1/\bar{\lambda}_{LT}^2$ [1](6.3.2.3(1))
$= 1/(0.99843 + \sqrt{0.99843^2 - 0.75 \times 0.9761^2}))$
$= 0.6537$
Lateral-torsional reduction factor

[54] $k_c = 1/(1.33 - 0.33\psi)$ [3](6.6)
$= \psi = 0.5$
$= 0.85837$
k_c for a linear moment diagram (derived)

[55] $f = 1 - 0.5(1 - k_c)[1 - 2(\bar{\lambda}_{LT} - 0.8)^2] \le 1$ [1](6.3.2.3(2))
$= 1 - 0.5 \times (1 - 0.85837) \times [1 - 2 \times (0.9761 - 0.8)^2]$
$= 0.93358$
Modification factor f

[56] $\chi_{LT,mod} = \chi_{LT}/f \le 1, \le 1/\bar{\lambda}_{LT}^2$ (eq. 6.58) [1](6.3.2.3(2))
$= 0.6537 / 0.93358,\ 1/\bar{\lambda}_{LT}^2 = 1.0496$
$= 0.70022$
Modified lateral-torsional reduction factor

[57] $\chi_{LT} = \chi_{LT,mod}$ [1](6.3.2.3)
$=$ as above
$= 0.70022$
Lateral-torsional reduction factor used

[58] $Mb_{Rd} = \chi_{LT} \cdot W_y \cdot f_y / \gamma_{M1}$ [1](6.3.2.1(3))
$= 0.70022 \times 2232.5\ cm^3 \times 345\ MPa / 1$
$= 539.3\ kN \cdot m$
Design buckling resistance moment
[59] $u_{LTB} = M_{y,Ed} / Mb_{Rd}$
$= 340\ kN \cdot m / 539.3\ kN \cdot m$
$= 0.63044$
Lateral-torsional buckling utilisation
[62] α_y = imperfection factor, curve a [1](6.3.1.2, Table 6.2)
= Table 6.2
= 0.21
Imperfection factor about y
[63] α_z = imperfection factor, curve b [1](6.3.1.2, Table 6.2)
= Table 6.2
= 0.34
Imperfection factor about z
[64] $N_{Rk} = A \cdot f_y$ [1](6.3.1.1(3))
$= 125.27\ cm^2 \times 345\ MPa$
$= 4321.6\ kN$
Characteristic axial resistance
[65] $N_{cr,y} = \pi^2 E \cdot I_y / L_{cr,y}^2$ [1](6.3.1.2)
$= \pi^2 \times 210000 \times 457280000 / 5000^2$
$= 37911\ kN$
Elastic critical force about y
[66] $N_{cr,z} = \pi^2 E \cdot I_z / L_{cr,z}^2$ [1](6.3.1.2)
$= \pi^2 \times 210000 \times 23470000 / 5000^2$
$= 1945.8\ kN$
Elastic critical force about z
[67] $\lambda_y = \sqrt{(A \cdot f_y / N_{cr,y})}$ [1](6.3.1.3)
$= \sqrt{(4321.6 / 37911)}$
$= 0.33763$
Non-dimensional slenderness about y
[68] $\lambda_z = \sqrt{(A \cdot f_y / N_{cr,z})}$ [1](6.3.1.3)
$= \sqrt{(4321.6 / 1945.8)}$
$= 1.4903$
Non-dimensional slenderness about z
[69] $\chi_y = 1 / (\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \leq 1$ [1](6.3.1.2)
$= \alpha = 0.21,\ \bar{\lambda} = 0.33763$
$= 0.96853$
Reduction factor about y
[70] $\chi_z = 1 / (\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \leq 1$ [1](6.3.1.2)
$= \alpha = 0.34,\ \bar{\lambda} = 1.4903$
$= 0.34582$
Reduction factor about z
[71] $N_{b,y,Rd} = \chi_y \cdot A \cdot f_y / \gamma_{M1}$ [1](6.3.1.1(3))
$= 0.96853 \times 4321.6 / 1$
$= 4185.7\ kN$
Buckling resistance about y
[72] $N_{b,z,Rd} = \chi_z \cdot A \cdot f_y / \gamma_{M1}$ [1](6.3.1.1(3))
$= 0.34582 \times 4321.6 / 1$
$= 1494.5\ kN$
Buckling resistance about z

[73] $N_{b,Rd} = \min(N_{b,y,Rd}, N_{b,z,Rd})$
$= \min(4185.7\ kN, 1494.5\ kN)$
$= 1494.5\ kN$
Flexural buckling resistance (smaller of the two axes)
[74] $M_{y,Rk} = W_y \cdot f_y$ [1](6.3.3)
$= 2232.5\ cm^3 \times 345\ MPa$
$= 770.2\ kN \cdot m$
Characteristic bending resistance about y
[75] $M_{z,Rk} = W_z \cdot f_y$ ($W_{pl,z}$ class 1, 2; $W_{el,z}$ class 3) [1](6.3.3)
$= 378.9\ cm^3 \times 345\ MPa$
$= 130.72\ kN \cdot m$
Characteristic bending resistance about z
[76] $C_{my} = 0.6 + 0.4\psi_y \geq 0.4$ (or entered) [3](B.3)
$= \psi_y = 0.5$
$= 0.8$
Equivalent uniform moment factor, y
[77] $C_{mz} = 0.6 + 0.4\psi_z \geq 0.4$ (or entered) [3](B.3)
$= \psi_z = 1$
$= 1$
Equivalent uniform moment factor, z
[78] C_{mLT} = as C_{my} (or entered) [3](B.3)
$= C_{my}$
$= 0.8$
Equivalent uniform moment factor for lateral-torsional buckling
[79] k_{yy} = Annex B: $C_{my}(1 + (\bar{\lambda}_y - 0.2)n_y) \leq C_{my}(1 + 0.8n_y)$ (class 1, 2) [4](B.2)
$= n_y = 0.095565$
$= 0.81052$
Interaction factor k_{yy}
[80] k_{yz} = Annex B: $0.6k_{zz}$ (class 1, 2), k_{zz} (class 3) [4](B.2)
$=$ as above
$= 0.82482$
Interaction factor k_{yz}
[81] k_{zy} = Annex B: Table B.1 $0.6k_{yy}$ / $0.8k_{yy}$; Table B.2 from $\bar{\lambda}_z$, n_z and C_{mLT} [4](B.2)
$= n_z = 0.26764$
$= 0.95134$
Interaction factor k_{zy}
[82] k_{zz} = Annex B: $C_{mz}(1 + (2\bar{\lambda}_z - 0.6)n_z) \leq C_{mz}(1 + 1.4n_z)$ (class 1, 2) [4](B.2)
$=$ as above
$= 1.3747$
Interaction factor k_{zz}
[83] $u_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$ [1](6.3.3(4))
$= 400\ kN / (0.96853 \times 4321.6\ kN / 1)$
$= 0.095565$
Axial term about y
[84] $u_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$ [1](6.3.3(4))
$= 400\ kN / (0.34582 \times 4321.6\ kN / 1)$
$= 0.26764$
Axial term about z
[85] $u_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$ [1](6.3.3(4))
$= 340\ kN \cdot m / (0.70022 \times 770.2\ kN \cdot m / 1)$
$= 0.63044$
Major-axis bending term

[86] $u_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$ [1](6.3.3(4))
$= 6\ kN{\cdot}m\ /\ (130.72\ kN{\cdot}m\ /\ 1)$
$= 0.045899$
Minor-axis bending term
[87] $u_{61} = N_{Ed}/(\chi_y\ N_{Rk}/\gamma_{M1}) + k_{yy}{\cdot}M_{y,Ed}/(\chi_{LT}\ M_{y,Rk}/\gamma_{M1}) + k_{yz}{\cdot}M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$ [1](6.3.3(4))
$= 0.095565 + 0.81052 \times 0.63044 + 0.82482 \times 0.045899$
$= 0.64441$
Interaction, equation 6.61
[88] $u_{62} = N_{Ed}/(\chi_z\ N_{Rk}/\gamma_{M1}) + k_{zy}{\cdot}M_{y,Ed}/(\chi_{LT}\ M_{y,Rk}/\gamma_{M1}) + k_{zz}{\cdot}M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$ [1](6.3.3(4))
$= 0.26764 + 0.95134 \times 0.63044 + 1.3747 \times 0.045899$
$= 0.9305$
Interaction, equation 6.62
[89] $u_{interaction} = \max(u_{61},\ u_{62})$ [1](6.3.3(4))
$= \max(0.64441,\ 0.9305)$
$= 0.9305$
Greatest interaction ratio
[90] $u_{buckling} = N_{Ed}/N_{b,Rd}$ [1](6.3.1.1)
$= 400\ kN\ /\ 1494.5\ kN$
$= 0.26764$
Flexural buckling utilisation
[91] $u_{member} = \max(\max(u_{interaction},\ u_{LTB}),\ u_{buckling})$
$= \max(\max(0.9305,\ 0.63044),\ 0.26764)$
$= 0.9305$
Governing utilisation of the member

5. Results

SYMBOL	VALUE	DESCRIPTION
Mb_{Rd}	539.30 kN·m	$M_{b,Rd}$
Mcr	808.38 kN·m	M_{cr}
χ_{LT}	0.70022	χ_{LT}
λ_{LT}	0.9761	$\bar{\lambda}_{LT}$
$N_{b,Rd}$	1494.5 kN	$N_{b,Rd}$
u_{LTB}	0.63	LTB utilisation
$u_{interaction}$	0.93	Interaction utilisation
u_{member}	0.93	Member utilisation
$edition_{used}$	"EN 1993-1-1:2005+A1:2014"	EN 1993-1-1 edition
ψ_y	0.5	ψ of the M_y diagram
$M_{y,Ed}$	340.00 kN·m	Greatest design moment about y-y
C_1	1.31	C_1
N_{Rk}	4321.6 kN	N_{Rk}
$N_{cr,y}$	37911 kN	$N_{cr,y}$
$N_{cr,z}$	1945.8 kN	$N_{cr,z}$
λ_y	0.33763	$\bar{\lambda}_y$
λ_z	1.4903	$\bar{\lambda}_z$

SYMBOL	VALUE	DESCRIPTION
χ_y	0.96853	χ_y
χ_z	0.34582	χ_z
$N_{b,y,Rd}$	4185.7 kN	$N_{b,y,Rd}$
$N_{b,z,Rd}$	1494.5 kN	$N_{b,z,Rd}$
k_{yy}	0.81052	k_{yy}
k_{yz}	0.82482	k_{yz}
k_{zy}	0.95134	k_{zy}
k_{zz}	1.3747	k_{zz}
α_{LT}	0.49	α_{LT}
α_y	0.21	α_y
α_z	0.34	α_z
Φ_{LT}	0.99843	Φ_{LT}
$\chi_{LT,0}$	0.6537	χ_{LT} before the f modification
k_c	0.85837	k_c
f	0.93358	f
W_y	2232.5 cm³	W_y
$M_{y,Rk}$	770.20 kN·m	$M_{y,Rk}$
$M_{z,Rk}$	130.72 kN·m	$M_{z,Rk}$
C_{my}	0.8	C_{my}
C_{mz}	1	C_{mz}
C_{mLT}	0.8	C_{mLT}
governing	"interaction 6.62 (0.931)"	Governing check
$u_{buckling}$	0.27	Buckling utilisation
u_{61}	0.64	Equation 6.61
u_{62}	0.93	Equation 6.62 (governing)
$u_{N,y}$	0.10	Axial term, y
$u_{N,z}$	0.27	Axial term, z
$u_{M,y}$	0.63	Bending term, y
$u_{M,z}$	0.05	Bending term, z

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Interaction, equation 6.62	u_{62} = 0.93	u_{62} ≤ 1	0.93	[1](6.3.3(4))
PASS	Interaction, equation 6.61	u_{61} = 0.64	u_{61} ≤ 1	0.64	[1](6.3.3(4))

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Lateral-torsional buckling $M_{y,Ed} \le M_{b,Rd}$	$u_{LTB} = 0.63$	$u_{LTB} \le 1$	0.63	[1](6.3.2.1)
PASS	Flexural buckling $N_{Ed} \le N_{b,Rd}$	$u_{buckling} = 0.27$	$u_{buckling} \le 1$	0.27	[1](6.3.1.1)

7. Applicability

WARNING records: tabulated values not taken from a reference record are E, G, f_y (Material properties); γ_{M1} (EN 1993-1-1 National Annex factors); α_{a0}, α_a, α_b, α_c, α_d, λ_{LT0}, β_{LT} (Imperfection factors and 6.3.2.3 parameters); C_2 (C_1, C_2 and k_c for the moment diagram). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

`Template struct.ltb-interaction@1.1.4 Content hash 4a826570ffcf88e3 Engine calc-engine/1.0.0`
`Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0`

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
2. NCCI SN003.
3. Table.
4. Annex B Table.